

Ankit Ghosh

SIGNAL & IMAGE PROCESSING · MACHINE LEARNING · INDUSTRIAL AI · ACOUSTIC SENSING

Bendixensweg 6, 22307 Hamburg, Germany

☎ (+49) 176-452-95766 | ✉ ankit.ghosh@protonmail.com | 🎮 arcan-3 | 🌐 ankit-ghosh

"If you want to find the secrets of the universe, think in terms of energy, frequency and vibration." - Nikola Tesla

Summary

Machine learning engineer and researcher working on the recovery of structure from noisy physical measurements: acoustic, vibration, optical and time-series. My doctorate built aMP, an end-to-end deep learning pipeline that replaces manual staging of meiosis in live-cell confocal microscopy, combining segmentation-based motion stabilisation, ensemble object localisation, constrained tracking and state-vector sequence modelling. Earlier work was in underwater acoustics: channel estimation in a water column, chirp-modulated communication, and passive SONAR source separation. I now lead industrial AI at BATO GmbH, owning platform architecture, on-premise deployment and technical delivery to manufacturing customers. I have worked in research and industry in parallel since 2015, including in regulated environments where correctness is defined by a standard text and an audit trail rather than by a validation metric alone. Python and PyTorch daily; C++ in production at DNV and Scalian; German at B2.

Work Experience

BATO GmbH (spun out of Hummingbird Converting Solutions)

Hamburg, Germany

MACHINE LEARNING ENGINEER, INDUSTRIAL AI — Technical Lead

Feb. 2025 - Present

- Lead industrial AI development for printing and process manufacturing: predictive maintenance, anomaly detection, production optimisation and operator decision support.
- Built a production copilot for a six-colour rotogravure printing line, combining machine settings, quality parameters, sensor data and deterministic process knowledge into operator recommendations.
- Own the architecture of **BATO Verse**, a tenant-aware industrial knowledge platform (FastAPI, PostgreSQL/pgvector, Neo4j, TimescaleDB, MinIO, Docker).
- Built LLM-based retrieval workflows linking time-series anomalies to technical manuals and troubleshooting knowledge to identify probable fault sources and suspect sensors.
- Developed a multilingual AI interview system (speech recognition, text-to-speech, locally hosted LLMs) that captures expert operator knowledge as structured knowledge-graph content.
- Designed secure on-premise GPU architectures with read-only PLC/SCADA integration, local inference and isolated storage, with no external cloud dependency for production data.

Aidy Health

Munich, Germany

AI CONSULTANT (part-time engagement)

Jun. 2025 - Oct. 2025

- Advised on signal processing for WiFi **Channel State Information (CSI)** data, targeting accuracy and real-time sensing performance.
- Developed models to detect and classify objects under varying motion and material conditions for passive environmental sensing.
- Refined CSI processing pipelines for IoT and smart-environment applications.

Metric-space (AI division, spun out of PANDA GmbH)

Hamburg, Germany

MACHINE / DEEP LEARNING ENGINEER

Sep. 2021 - Sep. 2024

- Built and deployed **vibroacoustic predictive maintenance** (v1.0) for rotating machinery, from raw sensor data through model development to production release.
- Developed **real-time acoustic event detection** for vehicle tampering on Jetson Xavier NX, using wavelet-decomposition-based preprocessing to meet latency targets on edge hardware under variable ambient noise.
- Designed an unsupervised forecasting and clustering method combining Meta's **Encodec** with the **PHATE** algorithm, reducing inference latency on large-scale time-series and surfacing latent regime structure.
- Built an unsupervised clustering and labelling toolbox for **equine gait analysis**, classifying movement anomalies across multi-view video to identify lameness indicators.
- Deployed ensemble adaptive-threshold **anomaly detection on infrared imagery** using the CFLOW anomaly foundation model for real-time industrial inspection.
- Implemented an **LLM chatbot over time-series datasets** using OpenAI function calling, letting analysts query and interrogate data conversationally.

Scalian

Hamburg, Germany

DATABASE APPLICATION DEVELOPER (working student, alongside doctoral research)

Feb. 2020 - Feb. 2021

- Migrated an Airbus finance model from PowerPivot to Python, converting spreadsheet-embedded logic into a maintainable, version-controlled codebase, and maintained it in service.
- Contributed to an in-house ERP application (C++, ASP.NET).

DNV - Renewable Advisory

Hamburg, Germany

SOFTWARE DEVELOPER & SIGNAL ANALYST

Aug. 2017 - Feb. 2019

- Analysed **harmonics, inter-harmonics and high-frequency content** in raw grid measurements for synchronous grid operation, and extended flicker analysis to current recommendations.
- Improved over- and under-voltage detection algorithms against defined standard thresholds.
- Implemented several **IEC standards** as verified applications with automated reporting and plotting (C++, VB.NET).

Basler AG

Ahrensburg, Germany

MACHINE VISION PLACEMENT — *Fishing for Experience* university–industry programme

Nov. 2015 - Mar. 2016

- Designed, built and installed a registered-mark (passer-mark) inspection system for **colour shift control** on a printing line, including a timer interface synchronising mark detection to press motion.

Education

Universität Hamburg

Hamburg, Germany

PH.D. IN MACHINE LEARNING AND IMAGE PROCESSING

Jan. 2019 - Oct. 2024

- Grade: cum laude (1.7). Advisor: Prof. Dr. Arp Schnittger. Second examiner: Prof. Dr. Sören Laue. Funded by the **Human Frontier Science Program (HFSP)**.
- Thesis: *aMP: Deep learning framework to characterize meiosis progression timeline in tetraploid Arabidopsis thaliana*. Built an end-to-end pipeline replacing manual staging of meiosis from confocal live-cell imaging: anther segmentation and motion stabilisation, meiocyte localisation, tracking, and stage classification.
- **Segmentation and stabilisation** of anthers via affine transformation to hold the focal plane across frames — Dice 88.9%, IoU 80.8%, F1 88.9% on the held-out test set.
- **Meiocyte localisation** using an ensemble of EfficientDet backbones with non-maximum suppression, evaluated under the COCO protocol — precision 0.77, recall 0.79, F1 0.78.
- **Tracking** through a restricted-space motion model, associating detections across frames on normalised cross-correlation and IoU, reducing computation time against unconstrained association.
- Represented meiosis phases as **state vectors**, validated predicted sequences against a knowledge graph, and derived Z-normalised staging pathways to build a piece-wise meiotic timeline. Per-stage classification accuracy 76–91%.
- Applied the timeline to **tetraploid A. thaliana**: meiosis I comparable to diploid, meiosis II markedly prolonged; quantified temporal dynamics in tcx5;6 mutants and ATM insertions.
- Reused the pipeline's modules for further quantification tasks — DNA double-strand-break foci, BiFC analysis, pollen counting.
- Continued as **postdoctoral researcher** (Oct. 2024 - Feb. 2025) on unsupervised staging of maize meiocytes.

Hamburg University of Technology (TUHH)

Hamburg, Germany

M.Sc. INFORMATION AND COMMUNICATION SYSTEMS

Oct. 2015 - Sep. 2018

- Overall grade 1.7. Major: communication systems.
- Master's thesis: *Source Separation and Echo Detection of Underwater Acoustic Signals of Whales* (grade 1.3), Helmut Schmidt Universität, Jan. - Jun. 2018, under Prof. Dr.-Ing. habil. Udo Zölzer.
- Project: chirp modulation for resilient underwater acoustic communication, Institute smartPORT, under Prof. Bernd-Christian Renner.
- Previously studied M.Sc. Microsystems at Bremen University of Applied Sciences (Oct. 2014 - Sep. 2015) before transferring; coursework average 1.4, project on **channel state estimation in a water column** under Prof. Dr. Dieter Kraus.

Institute of Engineering and Management

Kolkata, India

B.TECH. ELECTRONICS AND COMMUNICATION ENGINEERING

Aug. 2010 - May 2014

- DGP 8.3.

Skills

Programming	Python, C++, C, MATLAB
Machine Learning	PyTorch, TensorFlow, Keras, NumPy
Methods	Signal & time-series processing, acoustic signal processing, image segmentation & tracking, anomaly detection, foundation-model adaptation, unsupervised clustering
Data & Infrastructure	PostgreSQL/pgvector, Neo4j, TimescaleDB, InfluxDB, MinIO, ReductStore, FastAPI, Docker
Deployment	On-premise GPU architectures, edge inference (Jetson), local LLM hosting, read-only PLC/SCADA integration
Languages	English (Native/Bilingual), Bengali (Native), Hindi, German (B2)

Research Projects

- *Source Separation and Echo Detection of Underwater Acoustic Signals of Whales*. Grade 1.3. Supervisor: Carlos de Obaldía, M.Sc. Examiners: Prof. Dr.-Ing. habil. Udo Zölzer, Prof. Dr. Bernd-Christian Renner.
- Built a system to detect **echolocation clicks of toothed whales** and extract three features from them sufficient to identify species, working from passive acoustic monitoring recordings containing multiple species and ambient ship noise.
- Isolated clicks from whistles, creaks and other harmonic vocalisations using **generalised Wiener-filtering source separation** — non-negative matrix factorization, median filtering and kernel additive modelling — compared by peak-to-noise ratio and signal-to-distortion ratio.
- Derived **echo decay time** via Hilbert transform analysis, establishing that each click pulse carries superimposed pulses.
- Separated each pulse into **high- and low-energy components** to detect the inlying short-term pulses and identify the frequency bands they occupy, using continuous wavelet analysis.
- Estimated **click length** through an adapted correlation technique exploiting the band-limitation of each pulse.
- Modelled shallow-water channel behaviour with synthetic impulse responses to reason about echo content when the source location is unknown.

Research group smartPORT, Hamburg University of Technology

Hamburg, Germany

- *Chirp Modulation for Resilient Underwater Acoustic Communication*. Grade 2.0. Supervisor and first examiner: Prof. Dr. Bernd-Christian Renner.
- Replaced frequency shift keying with **chirp modulation** for packet-based preamble synchronisation between micro autonomous underwater vehicles (μ AUVs), where multipath propagation was making FSK preamble boundaries ambiguous.
- Built the signal generation algorithm and evaluation chain, and characterised the acoustic modem and Tiepie Handyscope HS5 instrumentation to select a workable parameter set.
- Validated across **four environments** of increasing realism — small container, test tank, Channel Harburg, and Außenmühle Lake — rather than in simulation alone.
- Showed chirp yields a prominent correlation peak where FSK produces a multipath-distorted triangular region, reducing preamble detection delay from **seconds to milliseconds** across ten iterations, with robustness to both multipath and Doppler shift.
- Verified the proportionality between **time-bandwidth product** and main-lobe power concentration, and established that the generation sampling rate could be reduced from 2 MHz to 500 kHz before spurious correlation peaks appeared — the constraint for porting to embedded hardware.

Selected Activities

Nanotechnology Solutions in Health — NanoMed concept study

Barcelona, Spain / Bydgoszcz, Poland

- Feasibility study for continuous monitoring of cancer treatment response: surveyed the enabling technologies and proposed a concept architecture of implantable biosensor motes, a dissolvable relay node, a wearable gateway and cloud reporting to clinicians.
- Assessed candidate biosensor modalities against measurable parameters, reviewed WBAN and Bluetooth as the in-body link, and mapped the EU medical device route to market (CE marking, MDD, AIMD, clinical phases).
- Built the business model canvas for the concept. Supervised by Mehmet Şükrü Kuran and Tuna Tugcu; judged best project among the participating international teams.

Institute of Engineering and Management

Kolkata, India

- *Pongopal*: Arduino-based floor-cleaning robot evaluating real-time path minimisation approaches. Grade 10/10.

Publications & Presentations

aMP: Deep learning framework to characterize meiosis progression timeline in tetraploid *Arabidopsis thaliana*

Universität Hamburg

- Published open access. URN `urn:nbn:de:gbv:18-ediss-123289` — `ediss.sub.uni-hamburg.de/handle/ediss/11301`

Microtubules: From Atoms to Complex Systems

Heidelberg, Germany

- Microtubule structures revealed as the main feature during deep-learning-based auto-staging of plant meiosis.

Plant Meiosis Meeting

Poznań, Poland

- Automated clustering techniques for meiosis staging in *Arabidopsis thaliana*.